

M004 Week 6 — English Worksheet

Topic: A Space that Reacts — Event → Check → Action · **Course:** Functions & Games · **Time:** about 45 minutes

This week your world **reacts to the player**. The pattern is always the same: an **event** happens (the player walks somewhere) → the code **checks** something (what block is there?) → an **action** runs (a door opens, a message appears, a trap triggers).

1 · Predict

Read each set of steps. Before you imagine the player doing it, write what you think will happen.

```
on player walks on gold block:
    say "Welcome to my world!"
```

When does the message appear? What happens when the player walks on grass instead?


```
on player walks on stone bricks:
    if block ahead is iron door:
        open the door
    otherwise:
        say "No door here."
```

The player walks on stone bricks right in front of the iron door. What happens? What happens on stone bricks far away from the door?


```
function spring_trap():  
    replace block below with air  
    say "You fell in my trap!"  
  
on player walks on red carpet:  
    spring_trap()
```

What happens when the player steps on red carpet? Why is the trap written as a function?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The door should open **only after** the code checks that there really is an iron door ahead. Right now the check and the action are **swapped**.

```
on player walks on stone bricks:  
    open the door  
    if block ahead is iron door:  
        say "Door found!"
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The trap is supposed to fire when the player steps on **red carpet**. But in the world, the red carpet does nothing and the blue carpet — the safe path — keeps eating players.

```
on player walks on blue carpet:  
    spring_trap()
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The secret message should appear **only** when the player stands on a gold block. Right now the message appears everywhere, on every single step.

```
on player walks:  
    say "You found the secret room!"
```

Hint: the event fires on every step — the **check** is missing.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now switch to your homework world. Build a space that reacts to the player — use the event → check → action pattern at least once. When you finish, come back here.

Send a photo or video of your space reacting, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

My space reacts when the player ...

The event is ... and the check is ...

The action that runs is ...

One tricky moment was when ...

To fix it, I ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while you walk through your reacting space. Talk through it like you are teaching someone who has never seen it. Try to use these words: **event**, **check**, **action**, **sensing**, **react**.

1. Show your space before anything happens, then start walking.
2. Walk into the spot that triggers the reaction and say which **event** fired.
3. Show one spot where **nothing** happens because the check is not true there.
4. Say in your own words what event → check → action means.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.